LESSON 5.7

Determine Surface Area of Prisms

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the surface area of rectangular and triangular prisms.

WHY THIS MATTERS Your class is painting the outside of different time capsule shapes to protect them from weather. Some capsules are rectangular prisms and others are triangular prisms (wedge-shaped). To buy the right amount of paint, you need to calculate the surface area of each shape! One capsule is a rectangular prism 7 inches long, 4 inches wide, and 3 inches tall. Another is a triangular prism whose triangle bases are 8 inches wide and 3 inches tall, with slant sides of 5 inches, and the prism is 9 inches long.

REMEMBER

Determine Surface Area of Prisms

Surface Area of Prisms. Formula: Surface Area = Sum of areas of all 6 faces Surface area is the total area of all the faces of a solid. A rectangular prism has 6 faces; a triangular prism has 5 faces (2 triangles + 3 rectangles). Add every face to get the total.

1. Calculate area of Top & Bottom: 2 × (l · w).
2. Calculate area of Front & Back: 2 × (l · h).
3. Calculate area of Left & Right: 2 × (w · h), then sum all three pairs

Words: **Surface area** ***(Área de superficie)*** **Rectangular prism** ***(Prisma rectangular)*** **Triangular prism** ***(Prisma triangular)*** **Face** ***(Cara)*** **Net** ***(Plantilla (desarrollo plano))***

1 **Circle the letter of the best answer. Show how you know.**

Surface area is measured in which type of unit?

(A) Cubic units (in³, cm³, ft³)
(B) Square units (in², cm², ft²)
(C) Linear units (in, cm, ft)
(D) No units are needed

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

How many faces does a rectangular prism have?

(A) 4
(B) 6
(C) 8
(D) 12

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A net of a rectangular prism shows six rectangles with areas 12, 12, 18, 18, 24 and 24 square inches. What is the surface area?

(A) 54 in²
(B) 72 in²
(C) 108 in²
(D) 108 in³

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

A net for a triangular prism has how many triangles and how many rectangles?

(A) 1 triangle and 4 rectangles
(B) 2 triangles and 3 rectangles
(C) 2 triangles and 4 rectangles
(D) 3 triangles and 2 rectangles

SHOW YOUR WORK

5 Circle the letter of the best answer. Show how you know.

A carpenter builds wooden storage chests. A rectangular chest is 3 feet long, 2 feet wide, and 2 feet tall. Wood stain covers 50 square feet per can. What is the chest's surface area?

(A) 12 ft²
(B) 16 ft²
(C) 24 ft²
(D) 32 ft²

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

6 Circle the letter of the best answer. Show how you know.

A carpenter builds wooden storage chests. A rectangular chest is 3 feet long, 2 feet wide, and 2 feet tall. Wood stain covers 50 square feet per can. How many cans of stain are needed for one chest?

(A) 0
(B) 1
(C) 2
(D) 3

PER 1		

1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

EXPLAIN YOUR THINKING

Compare Capsule A (rectangular, SA = 248 in²) and Capsule C (triangular, SA = 184 in²). Even though they are similar in size, why does the rectangular prism need more paint?

SENTENCE STARTER The rectangular prism has ___ faces while the triangular prism has ___ faces. The rectangular prism's SA is ___ in² which is ___ more than the triangular prism's ___ in² because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? □ 4 · I can teach it □ 3 · I've got it □ 2 · I need a hint □ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Find each prism's surface area. Then sort it by whether it needs 96 in² of paper or less, or more than 96 in².

Groups: 96 in² of paper or less | More than 96 in²

_____ Rect capsule 3×4×5

_____ Cube capsule 4×4×4

_____ Rect capsule 6×5×4

_____ Tri capsule (ends 8×6, length 10)

2 **Write the letter of the matching item on each line.**

Match each rectangular capsule's dimensions to how much cardboard wraps its surface.

_____ 7 in × 3 in × 2 in

_____ 6 in × 4 in × 2 in

_____ 9 in × 2 in × 2 in

_____ 6 in × 5 in × 3 in

(A) 82 in²

(B) 88 in²

(C) 80 in²

(D) 126 in²

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A carpenter builds wooden storage chests. A rectangular chest is 3 feet long, 2 feet wide, and 2 feet tall. Wood stain covers 50 square feet per can. The carpenter builds a second identical chest. Now how many cans?

(A) 1
(B) 2
(C) 3
(D) 4

PER 1

4 **Circle the letter of the best answer. Show how you know.**

A rectangular time capsule box measures 3 in by 4 in by 5 in. How much wrapping paper covers its surface (no overlap)?

(A) 47 in²
(B) 60 in²
(C) 94 in²
(D) 94 in³

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A rectangular capsule crate is 6 cm long, 5 cm wide, and 4 cm tall. How much cardboard is needed to cover all its faces?

(A) 74 cm²
(B) 120 cm²
(C) 148 cm²
(D) 148 cm³

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A triangular prism is 10 in long. Its triangle ends have base 8 in, height 6 in, and sides 8, 6, and 10 in. Find its surface area.

(A) 48 in²
(B) 240 in²
(C) 288 in²
(D) 288 in³

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

EXPLAIN YOUR THINKING

Compare Capsule A (rectangular, SA = 248 in²) and Capsule C (triangular, SA = 184 in²). Even though they are similar in size, why does the rectangular prism need more paint?

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